

PGR106

Linear Algebra

Lecture 8 – Eigenvalues & Eigenvectors



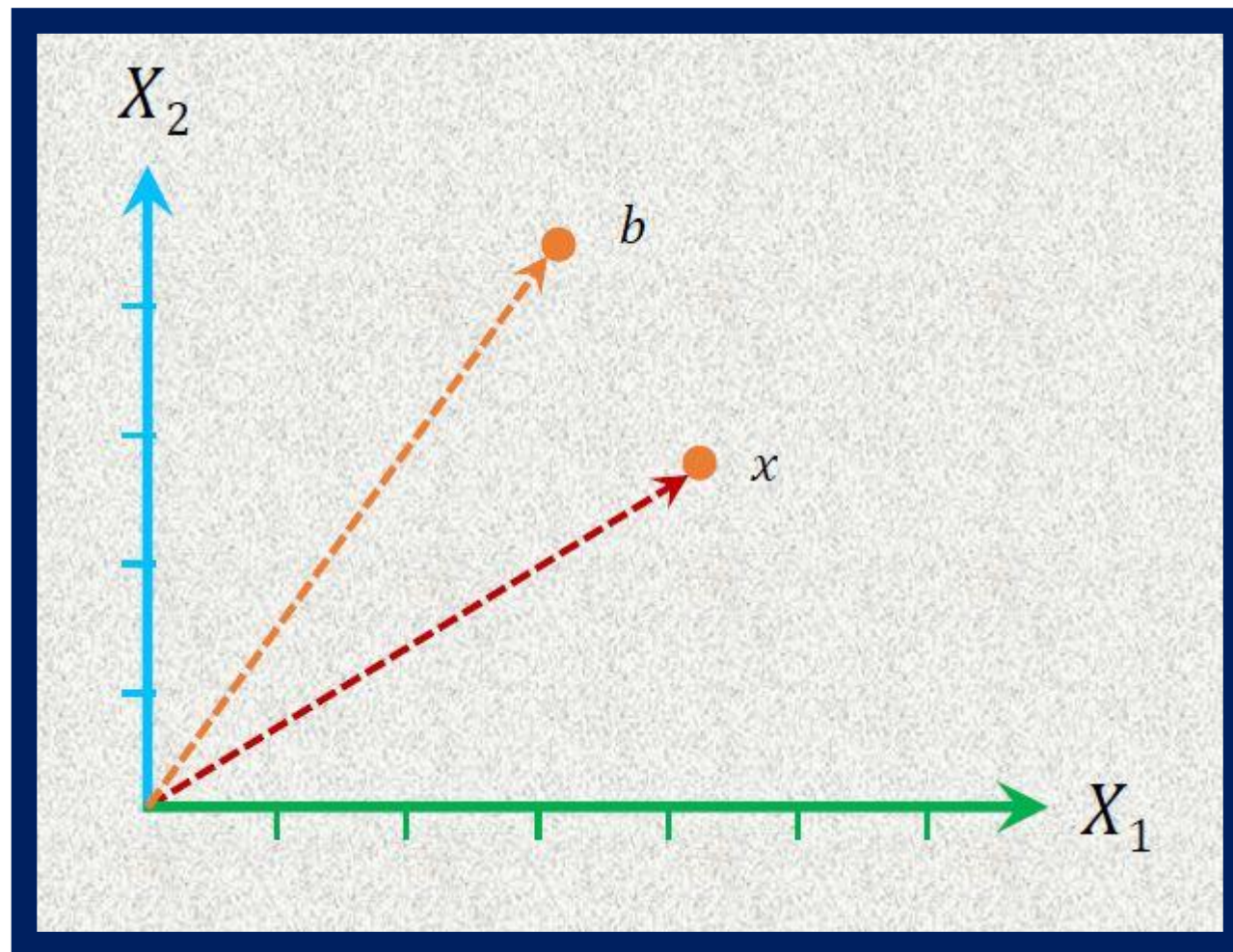
Eigenvalues & Eigenvectors

- German word “Eigen” -> “Characteristic” in English
- Eigenvalues and eigenvectors help in understating characteristics of linear transformations.
- Applications:
 - ✓ Solving Linear Equations
 - ✓ Singular Value Decomposition
 - ✓ Compression of Data
 - ✓ Principle Component Analysis

Eigenvalues & Eigenvectors

- Matrix-vector product equation: $Ax = b$
- Geometric interpretation of matrix-vector product:

When vector x is operated by A , a new vector b with a different orientation is obtained.



Eigenvalues & Eigenvectors

- Can we find a vector $\mathbf{b}$ with the same orientation as vector $\mathbf{x}$?

$$A\mathbf{x} = \mathbf{b}$$

- Then, the vector $\mathbf{b}$ will be a scalar multiple of vector $\mathbf{x}$.

$$\mathbf{b} = \lambda \mathbf{x}$$

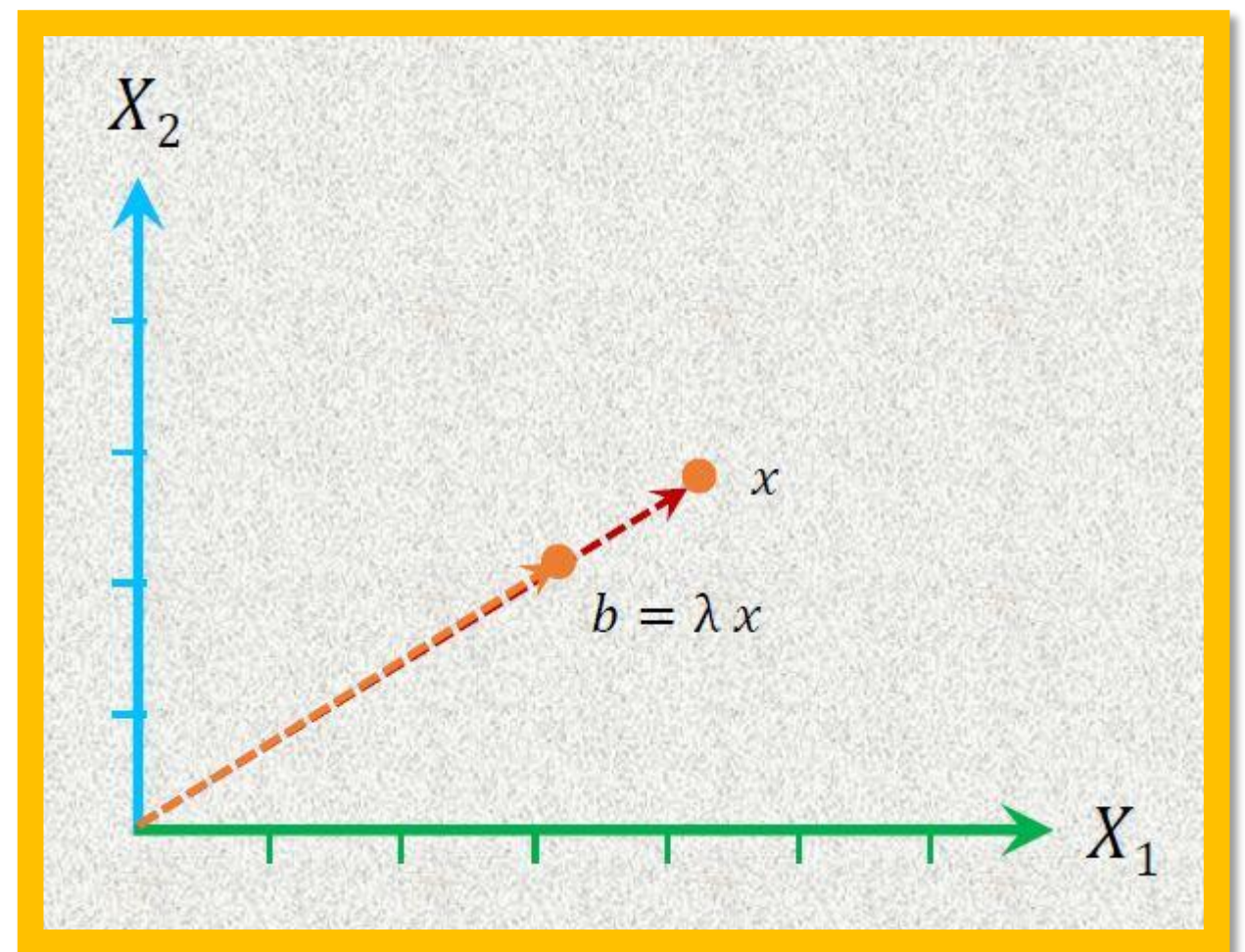
$$A\mathbf{x} = \mathbf{b} = \lambda \mathbf{x}$$

$$n \times n \rightarrow A\mathbf{x} = \lambda \mathbf{x}$$

Eigenvalue

Eigenvector

$$n \times 1$$

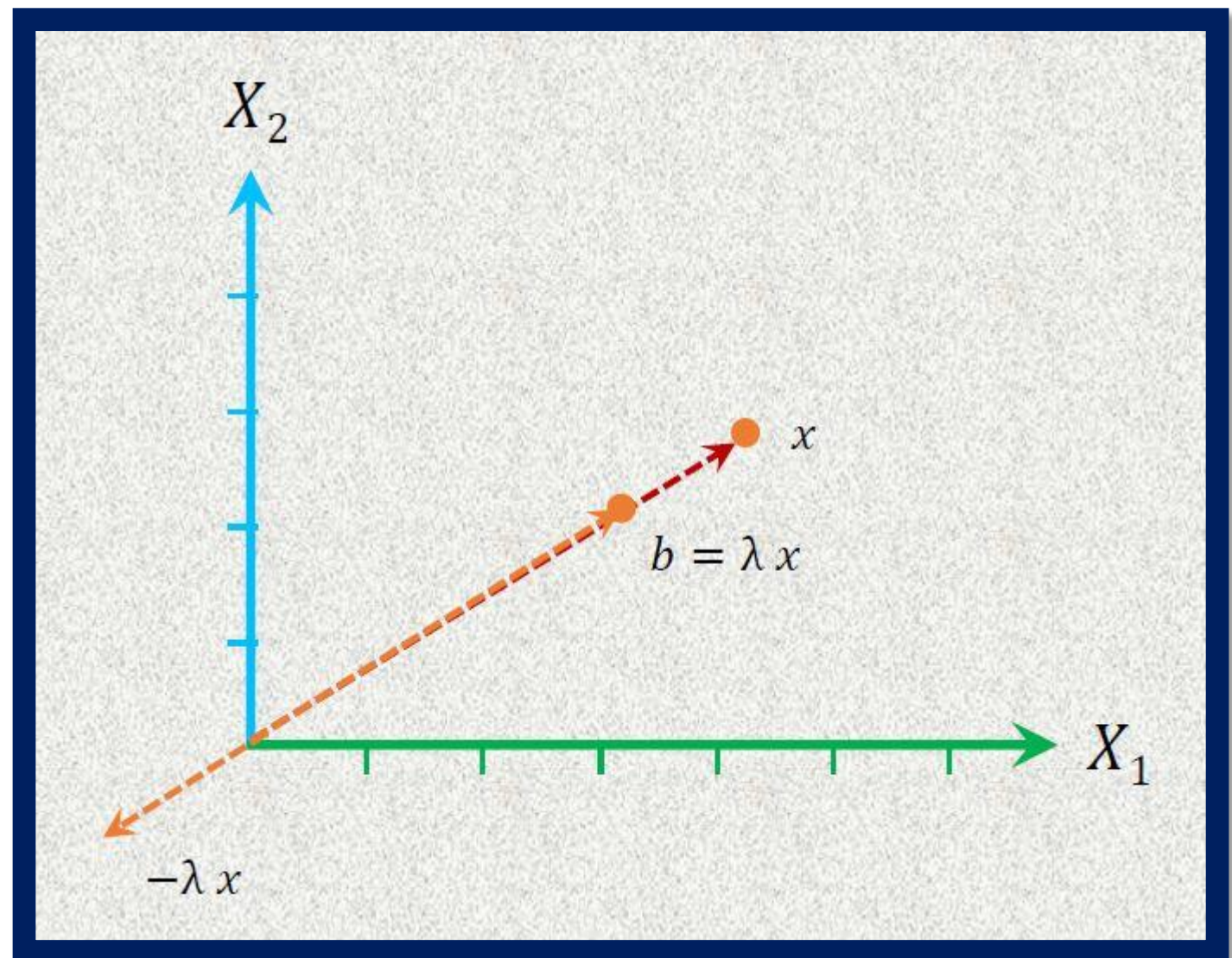


Eigenvalues & Eigenvectors

- Eigenvalue or scalar value λ can be positive or negative.

$$Ax = \lambda x$$

Eigenvalue



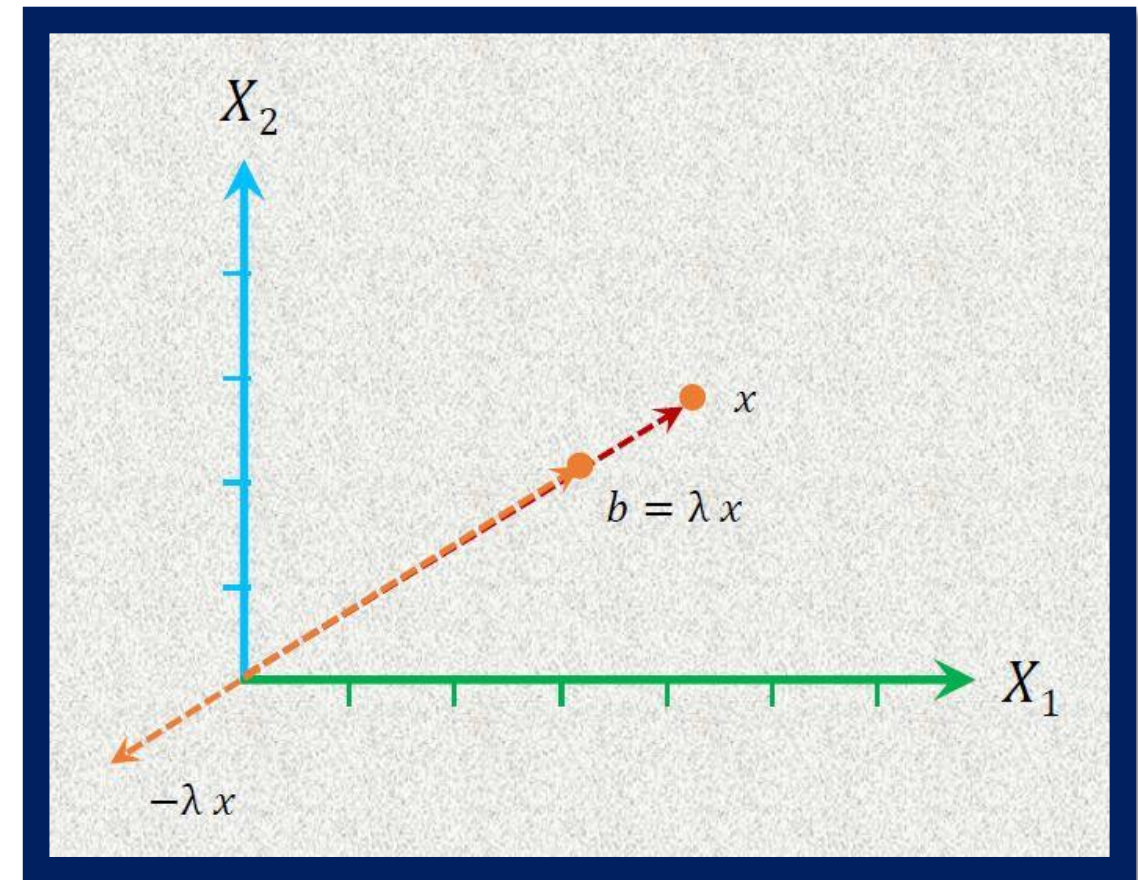
Positive λ : Amount of stretch or shrinkage in x direction

Negative λ : Amount of stretch in opposite direction of x

Eigenvalues & Eigenvectors

$$A\mathbf{x} = \lambda\mathbf{x}$$

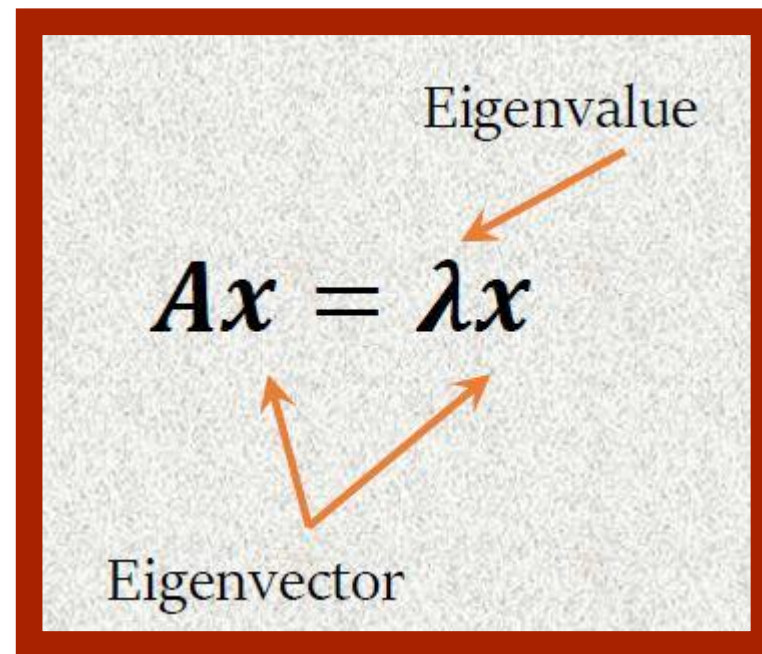
Eigenvalue
↙



Suppose Eigenvalue:

- $\lambda = 1$ No change in vector $\mathbf{x}$
- $\lambda = 2$ Double the length of vector $\mathbf{x}$
- $\lambda = -1$ No change in length of vector $\mathbf{x}$ but reverse direction
- $\lambda = -2$ Double the length of vector $\mathbf{x}$ also in reverse direction

Eigenvalues & Eigenvectors



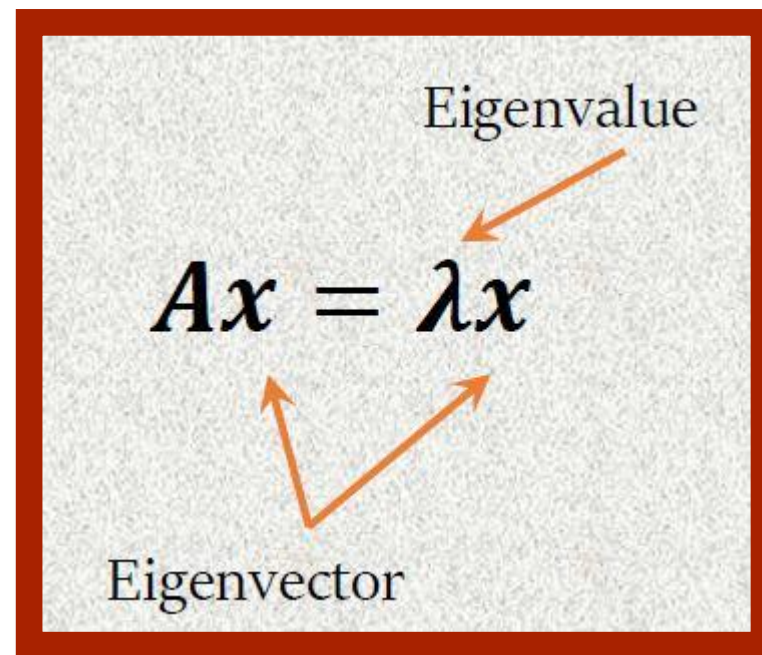
A diagram showing the equation $Ax = \lambda x$ inside a red-bordered box. An orange arrow points from the word "Eigenvalue" to the symbol λ . Two orange arrows point from the word "Eigenvector" to the symbols x and Ax .

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \quad x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 2 * 1 + 5 * 1 \\ 6 * 1 + 1 * 1 \end{bmatrix} = \begin{bmatrix} 2 + 5 \\ 6 + 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix} = 7 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 7x$$

Eigenvalues & Eigenvectors



A diagram showing the equation $Ax = \lambda x$ inside a red-bordered box. An orange arrow points from the word "Eigenvalue" to the symbol λ . Two orange arrows point from the word "Eigenvector" to the x terms on both sides of the equation.

$$Ax = \lambda x$$

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \quad x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 2 * 2 + 5 * 3 \\ 6 * 2 + 1 * 3 \end{bmatrix} = \begin{bmatrix} 4 + 15 \\ 12 + 3 \end{bmatrix} = \begin{bmatrix} 19 \\ 15 \end{bmatrix}$$

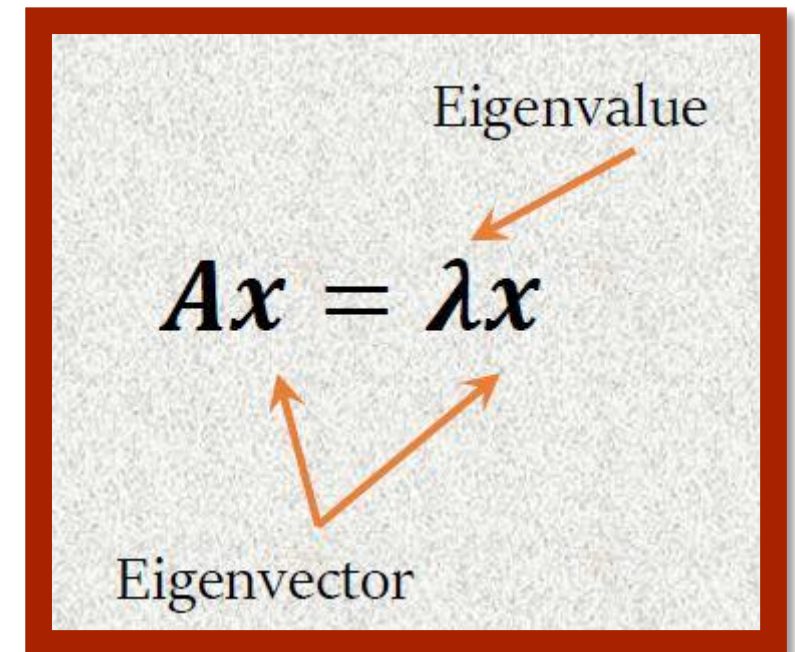
Eigenvalues & Eigenvectors

- Computing **eigenvectors** from **eigenvalues**

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \quad \lambda = -4$$

$$\begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -4 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 2x_1 + 5x_2 \\ 6x_1 + x_2 \end{bmatrix} = \begin{bmatrix} -4x_1 \\ -4x_2 \end{bmatrix}$$



A diagram illustrating the eigenvalue equation $Ax = \lambda x$. The equation is centered within a red-bordered box. An orange arrow points from the label "Eigenvalue" to the symbol λ . Another orange arrow points from the label "Eigenvector" to the vector x .

Eigenvalues & Eigenvectors

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \quad \lambda = -4$$

$$\begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -4 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 2x_1 + 5x_2 \\ 6x_1 + x_2 \end{bmatrix} = \begin{bmatrix} -4x_1 \\ -4x_2 \end{bmatrix}$$

$$\left. \begin{array}{l} 2x_1 + 5x_2 = -4x_1 \\ 6x_1 + x_2 = -4x_2 \end{array} \right\} \quad 6x_1 + 5x_2 = 0$$

Eigenvalues & Eigenvectors

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \quad \lambda = -4$$

$$6x_1 + 5x_2 = 0$$

$$x_1 = -5 \quad x_2 = 6$$

$$x = \begin{bmatrix} -5 \\ 6 \end{bmatrix}$$

Eigenvalues & Eigenvectors

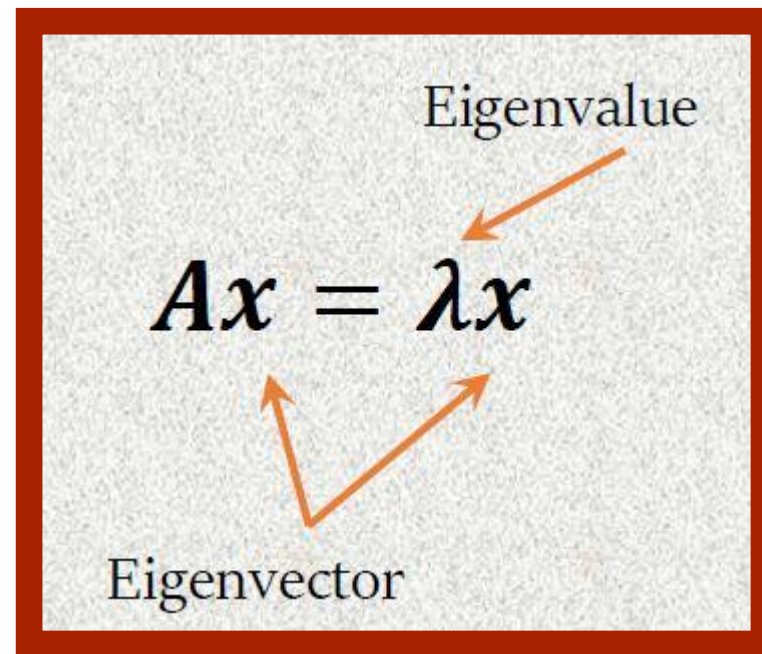
$$A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \quad \lambda = -4 \quad x = \begin{bmatrix} -5 \\ 6 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \begin{bmatrix} -5 \\ 6 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 2 * -5 + 5 * 6 \\ 6 * -5 + 1 * 6 \end{bmatrix} = \begin{bmatrix} -10 + 30 \\ -30 + 6 \end{bmatrix} = \begin{bmatrix} 20 \\ -24 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 20 \\ -24 \end{bmatrix} = -4 \begin{bmatrix} -5 \\ 6 \end{bmatrix}$$

Characteristic Equation


$$Ax = \lambda x$$

Eigenvalue

Eigenvector

$$Ax - \lambda x = 0$$

$$\begin{bmatrix} a_{1,1} & a_{1,2} & \cdot & \cdot & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdot & \cdot & a_{2,n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n,1} & a_{n,2} & \cdot & \cdot & a_{n,n} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} - \lambda \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}$$

$n \times n$ $n \times 1$ $n \times 1$ $n \times 1$

Characteristic Equation

$$Ax - \lambda x = 0$$

$$\begin{bmatrix} a_{1,1} & a_{1,2} & \cdot & \cdot & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdot & \cdot & a_{2,n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n,1} & a_{n,2} & \cdot & \cdot & a_{n,n} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} - \lambda \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}$$

$n \times n \qquad n \times 1 \qquad n \times 1 \qquad n \times 1$

$$\begin{bmatrix} a_{1,1} & a_{1,2} & \cdot & \cdot & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdot & \cdot & a_{2,n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n,1} & a_{n,2} & \cdot & \cdot & a_{n,n} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 & \cdot & 0 \\ 0 & 1 & 0 & \cdot & 0 \\ 0 & 0 & 1 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}$$

$n \times n \qquad n \times 1 \qquad n \times n \qquad n \times 1 \qquad n \times 1$

Characteristic Equation

$$Ax = \lambda x$$

$$\begin{bmatrix} a_{1,1} & a_{1,2} & \cdot & \cdot & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdot & \cdot & a_{2,n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n,1} & a_{n,2} & \cdot & \cdot & a_{n,n} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 & \cdot & 0 \\ 0 & 1 & 0 & \cdot & 0 \\ 0 & 0 & 1 & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}$$

$$Ax - \lambda Ix = 0$$

Characteristic Equation

$$Ax = \lambda x$$

$$Ax - \lambda Ix = 0$$

$$(A - \lambda I)x = 0$$

$$(A - \lambda I) = 0$$

$$x \neq 0$$

- **Eigenvalues** can be identified using:

$$|A - \lambda I| = 0$$

- Substituting eigenvalues in the equation will give **eigenvectors**.

Characteristic Equation

Example

$$A = \begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix}$$

$$Ax = \lambda x$$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$(A - \lambda I) x = 0$$

$$\left(\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$|A - \lambda I| = \left| \begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 8 - \lambda & 7 \\ 2 & 3 - \lambda \end{bmatrix} \right| = 0$$

$$(8 - \lambda)(3 - \lambda) - 2 * 7 = 0$$

$$24 - 8\lambda - 3\lambda + \lambda^2 - 14 = 0$$

$$\lambda^2 - 11\lambda + 10 = 0$$

$$\underline{\lambda = (10, 1)}$$

Characteristic Equation

Example

$$A = \begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix}$$

$$\lambda = 1$$



$$Ax = \lambda x$$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 1 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 8x_1 + 7x_2 \\ 2x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\left. \begin{array}{l} 8x_1 + 7x_2 = x_1 \\ 2x_1 + 3x_2 = x_2 \end{array} \right\} x_1 + x_2 = 0$$

$$\text{Magnitude of: } \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

Eigenvector:

Unit vector

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ 1 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$x_1 + x_2 = 0$$



$$x_1 = 1$$

$$x_2 = -1$$

Characteristic Equation

Example

$$A = \begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix}$$

$$\lambda = 10$$



$$Ax = \lambda x$$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 10 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 8x_1 + 7x_2 \\ 2x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} 10x_1 \\ 10x_2 \end{bmatrix}$$

$$\left. \begin{array}{l} 8x_1 + 7x_2 = 10x_1 \\ 2x_1 + 3x_2 = 10x_2 \end{array} \right\} 7x_2 = 2x_1$$

$$\text{Magnitude of: } \begin{bmatrix} 7 \\ 2 \end{bmatrix} = \sqrt{7^2 + 2^2} = \sqrt{49 + 4} = \sqrt{53}$$

Eigenvector:

Unit vector

$$\begin{bmatrix} \frac{7}{\sqrt{53}} \\ \frac{2}{\sqrt{53}} \end{bmatrix}$$

$$7x_2 = 2x_1 \longrightarrow$$

$$x_1 = 7$$

$$x_2 = 2$$

Eigen Decomposition

- **Eigen Decomposition** decomposes a matrix into **eigenvectors** and **eigenvalues**.

$$Ax = \lambda x$$



$$A = Q \Lambda Q^{-1}$$

Eigenvectors: $v_1, v_2, \dots, v_n$

Eigenvalues: $\lambda_1, \lambda_2, \dots, \lambda_n$

$$Q = [v_1 \ v_2 \ \dots \ v_n]$$

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Eigen Decomposition

$$A = Q \Lambda Q^{-1}$$

Eigenvectors: $v_1, v_2, \dots, v_n$

Eigenvalues: $\lambda_1, \lambda_2, \dots, \lambda_n$

$$A = [v_1 \ v_2 \ \dots \ v_n] \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix} [v_1 \ v_2 \ \dots \ v_n]^{-1}$$

Eigen Decomposition

Example

$$A = Q \Lambda Q^{-1}$$

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \quad \lambda_1 = 7 \quad v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \lambda_2 = -4 \quad v_2 = \begin{bmatrix} -5 \\ 6 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 7 & 0 \\ 0 & -4 \end{bmatrix} \quad Q^{-1} = \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}^{-1}$$
$$Q^{-1} = \frac{1}{11} \begin{bmatrix} 6 & 5 \\ -1 & 1 \end{bmatrix}$$
$$Q^{-1} = \begin{bmatrix} 6/11 & 5/11 \\ -1/11 & 1/11 \end{bmatrix}$$

Eigen Decomposition

Example

$$A = Q \Lambda Q^{-1}$$

$$Q = \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 7 & 0 \\ 0 & -4 \end{bmatrix}$$

$$Q^{-1} = \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}^{-1}$$

$$Q^{-1} = \frac{1}{11} \begin{bmatrix} 6 & 5 \\ -1 & 1 \end{bmatrix}$$

$$Q^{-1} = \begin{bmatrix} 6/11 & 5/11 \\ -1/11 & 1/11 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} 7 & 0 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} 6/11 & 5/11 \\ -1/11 & 1/11 \end{bmatrix}$$

Eigen Decomposition

Example

$$A = Q \Lambda Q^{-1}$$

$$A = \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} 7 & 0 \\ 0 & -4 \end{bmatrix} \begin{bmatrix} 6/11 & 5/11 \\ -1/11 & 1/11 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 * 7 + (-5) * 0 & 1 * 0 + (-5) * (-4) \\ 1 * 7 + 6 * 0 & 1 * 0 + 6 * (-4) \end{bmatrix} \begin{bmatrix} 6/11 & 5/11 \\ -1/11 & 1/11 \end{bmatrix}$$

$$A = \begin{bmatrix} 7 & 20 \\ 7 & -24 \end{bmatrix} \begin{bmatrix} 6/11 & 5/11 \\ -1/11 & 1/11 \end{bmatrix}$$

Eigen Decomposition

Example

$$A = Q \Lambda Q^{-1}$$

$$A = \begin{bmatrix} 7 * 6/11 + 20 * -1/11 & 7 * 5/11 + 20 * 1/11 \\ 7 * 6/11 + (-24) * -1/11 & 7 * 5/11 + (-24) * 1/11 \end{bmatrix}$$

$$A = \begin{bmatrix} 42/11 - 20/11 & 35/11 + 20/11 \\ 42/11 + 24/11 & 35/11 - 24/11 \end{bmatrix} = \begin{bmatrix} 22/11 & 55/11 \\ 66/11 & 11/11 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix}$$

Diagonalization

- **Diagonalization** of a matrix is to transform the matrix into a diagonal matrix.

$$A = Q \Lambda Q^{-1}$$

$$D = P^{-1} A P$$

Eigenvectors: $v_1, v_2, \dots, v_n$

$$P = \begin{bmatrix} v_1 & v_2 & \cdot & \cdot & v_n \end{bmatrix}$$

Diagonalization

Example

$$D = P^{-1} A P$$

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \quad \lambda_1 = 7 \quad v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \lambda_2 = -4 \quad v_2 = \begin{bmatrix} -5 \\ 6 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}^{-1}$$

$$P^{-1} = \frac{1}{11} \begin{bmatrix} 6 & 5 \\ -1 & 1 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} 6/11 & 5/11 \\ -1/11 & 1/11 \end{bmatrix}$$

Diagonalization

Example

$$D = P^{-1} A P$$

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix}$$

$$\lambda_1 = 7 \quad v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = -4 \quad v_2 = \begin{bmatrix} -5 \\ 6 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} 6/11 & 5/11 \\ -1/11 & 1/11 \end{bmatrix}$$

$$D = \begin{bmatrix} 6/11 & 5/11 \\ -1/11 & 1/11 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & -4 \end{bmatrix}$$

Diagonalization

Example

$$D = P^{-1} A P$$

$$D = \begin{bmatrix} 6/11 & 5/11 \\ -1/11 & 1/11 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix} \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}$$

$$D = \begin{bmatrix} 6/11 * 2 + 5/11 * 6 & 6/11 * 5 + 5/11 * 1 \\ -1/11 * 2 + 1/11 * 6 & -1/11 * 5 + 1/11 * 1 \end{bmatrix} \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}$$

$$D = \begin{bmatrix} 12/11 + 30/11 & 30/11 + 5/11 \\ -2/11 + 6/11 & -5/11 + 1/11 \end{bmatrix} \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}$$

Diagonalization

Example

$$D = P^{-1} A P$$

$$D = \begin{bmatrix} 12/11 + 30/11 & 30/11 + 5/11 \\ -2/11 + 6/11 & -5/11 + 1/11 \end{bmatrix} \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}$$

$$D = \begin{bmatrix} 42/11 & 35/11 \\ 4/11 & -4/11 \end{bmatrix} \begin{bmatrix} 1 & -5 \\ 1 & 6 \end{bmatrix}$$

$$D = \begin{bmatrix} 42/11 * 1 + 35/11 * 1 & 42/11 * -5 + 35/11 * 6 \\ 4/11 * 1 + -4/11 * 1 & 4/11 * -5 + -4/11 * 6 \end{bmatrix}$$

Diagonalization

Example

$$D = P^{-1} A P$$

$$D = \begin{bmatrix} 42/11 * 1 + 35/11 * 1 & 42/11 * -5 + 35/11 * 6 \\ 4/11 * 1 + -4/11 * 1 & 4/11 * -5 + -4/11 * 6 \end{bmatrix}$$

$$D = \begin{bmatrix} 42/11 + 35/11 & -210/11 + 210/11 \\ 4/11 + -4/11 & -20/11 - 24/11 \end{bmatrix}$$

$$D = \begin{bmatrix} 77/11 & 0 \\ 0 & -44/11 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & -4 \end{bmatrix}$$

Eigenvalues & Eigenvectors

Exercise

Consider the following matrix

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

Find its eigenvalues and eigenvectors, and verify the eigendecomposition of A . Check your solution in Python.

Eigenvalues & Eigenvectors

Exercise

Consider the following matrix

$$B = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

Find its eigenvalues and eigenvectors. Check your solution in Python.

Eigenvalues & Eigenvectors

Exercise

Diagonalize the following matrix

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Eigenvalues & Eigenvectors

Exercise

Diagonalize the following matrix

$$A = \begin{bmatrix} 6 & 2 & 1 \\ 2 & 6 & 2 \\ 1 & 2 & 6 \end{bmatrix}$$

End of Lecture 8



MATHEMATICS FOR MACHINE LEARNING

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